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LOAN APPROVAL PREDICTION

AI for Engineers (UCS321)

MST Mini Project,

Problem 5

1. Problem Statement and Objective

Banks must decide whether to approve or reject a loan application. This project builds a supervised machine learning model that predicts loan approval status (Approved / Rejected) for an HDFC-style bank, using applicant financial and demographic data. The task is framed as a binary classification problem: the model outputs one of two classes, not a continuous number, which is why classification algorithms (not regression) are used.

Objective: train and compare multiple classifiers, select the best one using proper evaluation metrics, and identify which applicant features most strongly drive the approval decision.

2. Dataset

Source: Kaggle-style loan approval dataset (loan_approval_dataset.csv), placed locally in data/.

Shape: 4,269 rows, 13 columns, no missing values, no duplicate rows.

Known quirk: column names and text values in the raw file had leading whitespace (e.g. " loan_status", " Approved"), which was stripped before any analysis.

Column Table

Column	Type	Meaning
loan_id	ID (dropped)	Unique row identifier, not predictive
no_of_dependents	Numeric	Number of dependents of the applicant
education	Categorical	Graduate / Not Graduate
self_employed	Categorical	Yes / No
income_annum	Numeric	Annual income (currency units)
loan_amount	Numeric	Requested loan amount
loan_term	Numeric	Loan term in years
cibil_score	Numeric	Credit score (300-900), India's standard credit score
residential_assets_value	Numeric	Value of residential assets owned
commercial_assets_value	Numeric	Value of commercial assets owned
luxury_assets_value	Numeric	Value of luxury assets owned
bank_asset_value	Numeric	Value of bank-held assets
loan_status	Target	Approved / Rejected -> mapped to 1 / 0

3. Pre-processing

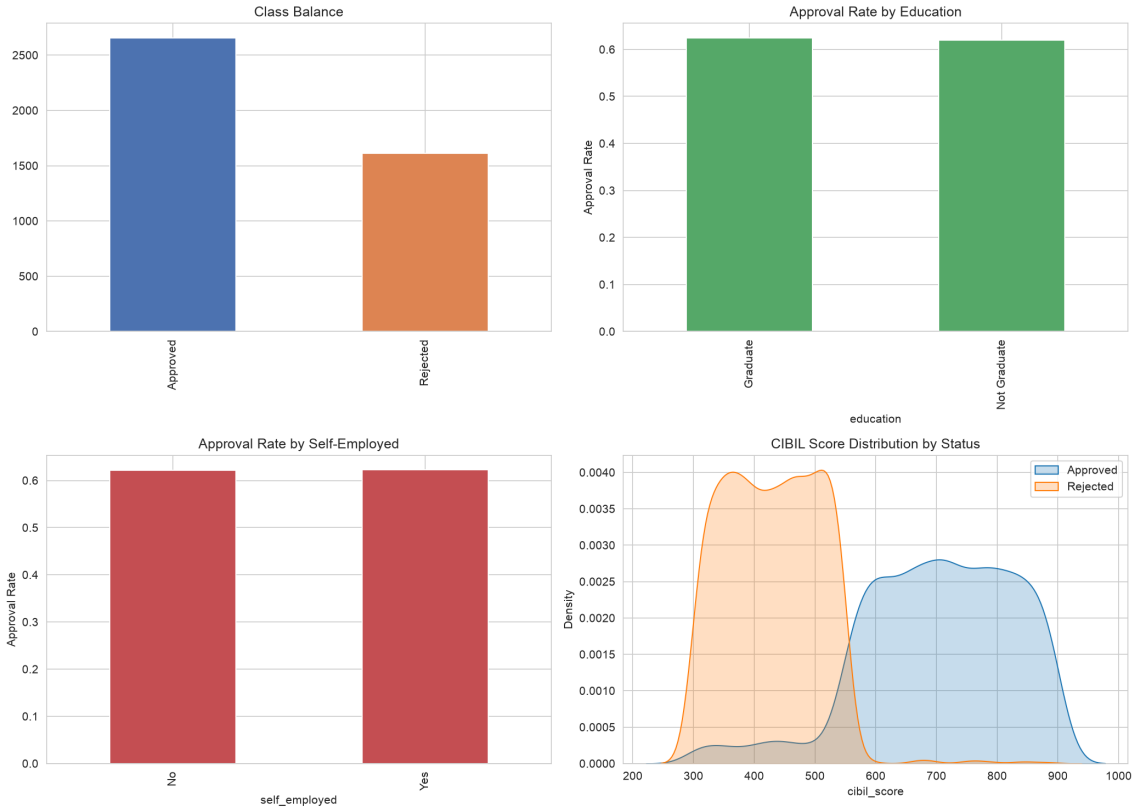
Issue	Fix	Reason
Leading whitespace in column names and text cells	df.columns.str.strip() and strip all string cells	Raw file quirk; would break equality checks like df["loan_status"]=="Approved"
loan_id column present	Dropped	Row identifier only, not predictive, would add noise
28 rows with negative residential_assets_value (min -100000)	Clipped to 0	Asset values cannot be negative in reality; treated as a data-entry error
No missing values found	No action needed	Verified via isna().sum(), confirmed zero across all columns
No duplicate rows found	No action needed	Verified via duplicated().sum(), confirmed zero
Categorical text columns (education, self_employed)	One-hot encoded with drop_first=True	Models need numeric input; drop_first avoids redundant columns
Features on very different scales (score 300-900 vs assets in millions)	StandardScaler inside a Pipeline, fit on train only	Prevents data leakage from test set into training; required for Logistic Regression to converge well
Mild class imbalance (62% Approved / 38% Rejected)	class_weight='balanced' on all models	Prevents models from being biased toward the majority class

Feature Engineering

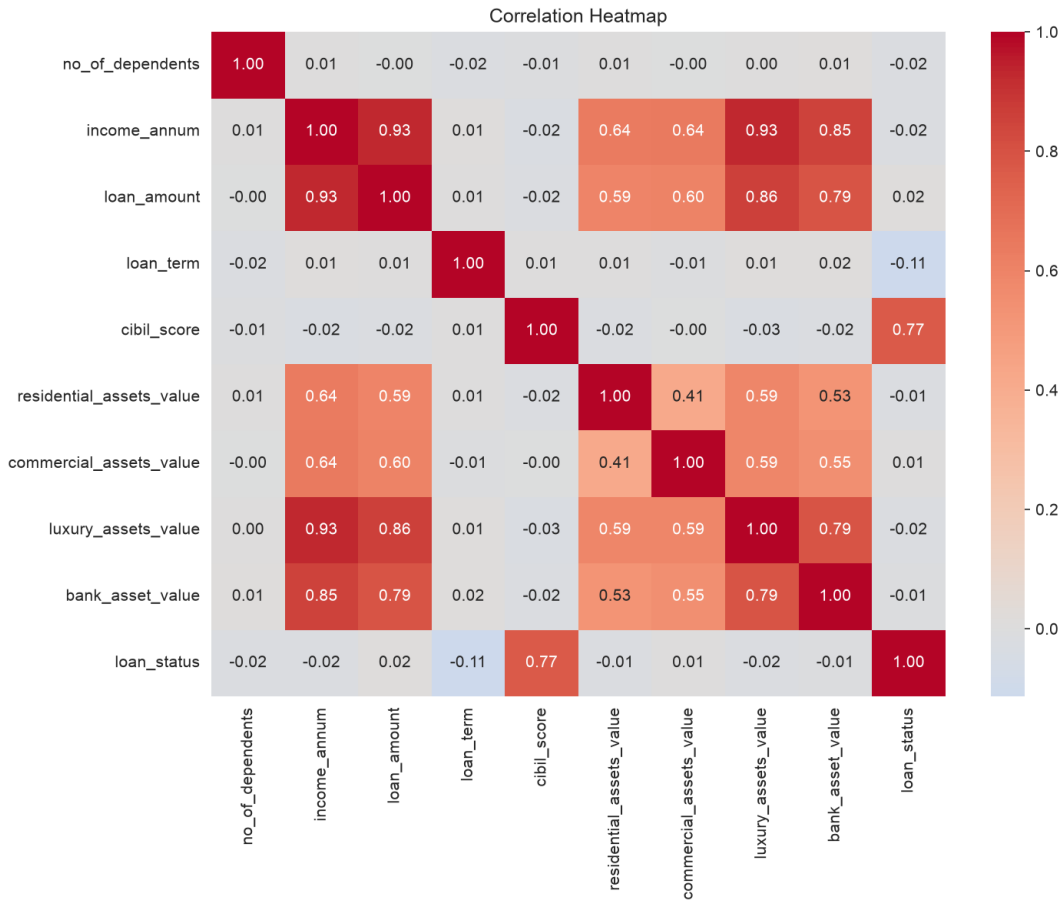
Four new features were built from the real columns:

- TotalAssets = residential + commercial + luxury + bank asset values
- Loan_to_Income = loan_amount / income_annum (higher = riskier applicant)
- Assets_to_Loan = TotalAssets / loan_amount (higher = more collateral, safer)
- EMI_style = loan_amount / loan_term (rough proxy for monthly repayment burden)

4. Exploratory Data Analysis (EDA)



Insight: Class balance is roughly 62% Approved / 38% Rejected. Approval rate is nearly identical across education and self-employment groups, meaning these are weak standalone predictors. CIBIL score distributions for Approved vs Rejected applicants barely overlap -- a strong early signal that credit score dominates the decision.



Insight: The correlation heat map confirms cibil_score has by far the strongest correlation with loan_status; other numeric features correlate weakly with the target but more with each other (e.g. the four asset columns).

5. Models

Three classifiers were trained, chosen to cover a spectrum of complexity and interpretability:

- Logistic Regression -- simple, interpretable linear baseline
- Decision Tree (max_depth=4) -- captures non-linear splits, easy to explain to a non-technical reviewer
- Random Forest (300 trees) -- ensemble of trees, usually more robust and less prone to overfitting than a single tree

All three were trained inside a scikit-learn Pipeline (StandardScaler -> classifier), with class_weight='balanced' to handle the mild class imbalance, and random_state=42 everywhere for reproducibility. The Random Forest was additionally tuned with GridSearchCV (5-fold CV, scoring=F1) over n_estimators, max_depth, and min_samples_split.

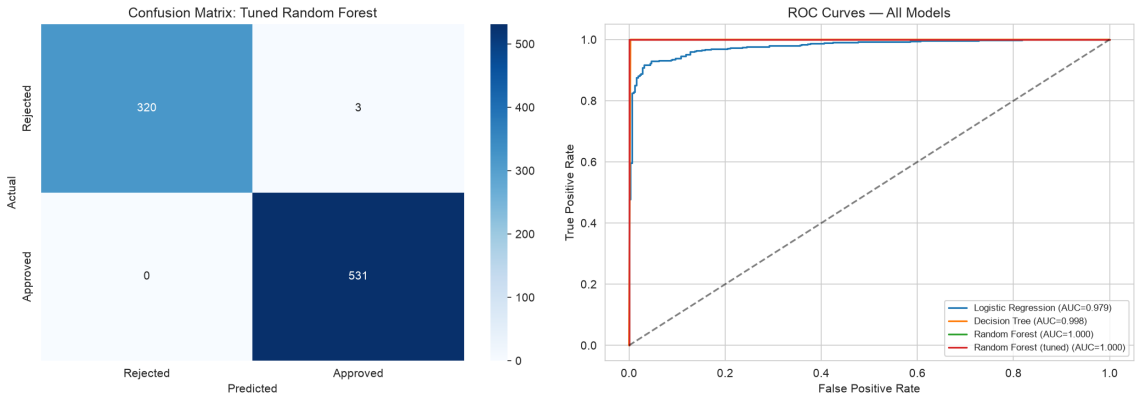
6. Results

Model	CV F1	Train Acc	Test Acc	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.9396	0.9283	0.9356	0.9704	0.9247	0.9470	0.9785
Decision Tree (depth=4	0.9993	1.0000	0.9988	0.9981	1.0000	0.9991	0.9985
Random Forest (300 trees)	0.9981	1.0000	0.9977	0.9962	1.0000	0.9981	1.0000
Random Forest (tuned)	0.9981	1.0000	0.9965	0.9944	1.0000	0.9972	1.0000

Best tuned Random Forest parameters: max_depth=None, min_samples_split=2, n_estimators=200.

Important honesty note: accuracy above 95%

All tree-based models exceed 95% test accuracy, which the project rules flag as a possible sign of data leakage. Investigation showed this is not a code bug: a single-feature rule, "cibil_score >= 550 -> Approved", already achieves 95.36% accuracy on the raw data. Approved applicants average a CIBIL score of 703.5, Rejected applicants average 429.5 -- the two groups barely overlap. This dataset appears to be synthetic or generated with a near-deterministic rule tied to credit score. The result is reported honestly rather than hidden, and its implications are discussed in Section 8 (Limits).

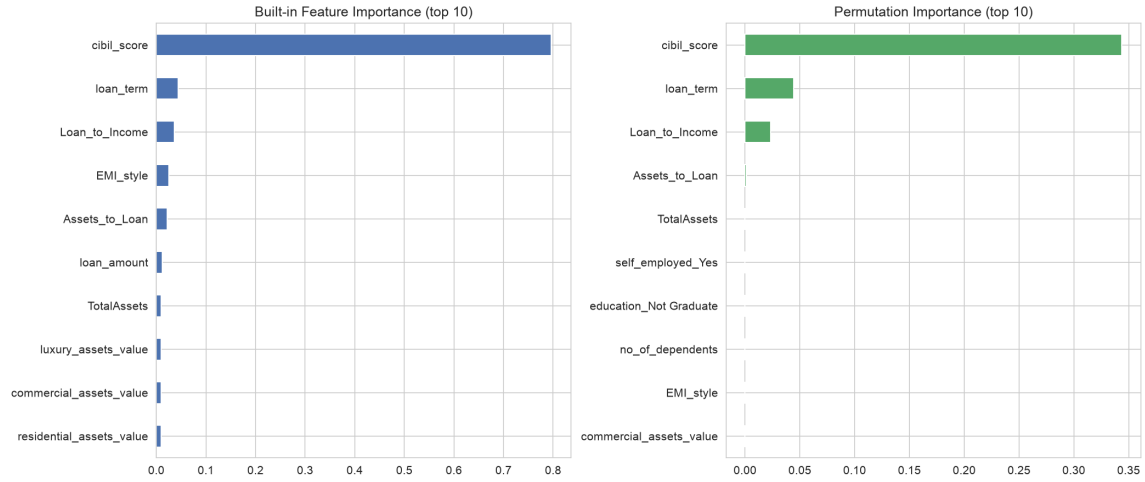


Confusion matrix (tuned Random Forest, test set, n=854): 320 correctly Rejected, 3 wrongly Approved, 0 wrongly Rejected, 531 correctly Approved. For a bank, a wrongly-Approved

applicant is the costlier error -- it risks the bank's money not being repaid -- while a wrongly-Rejected applicant only costs a missed business opportunity.

7. Innovation

- Engineered features: TotalAssets, Loan_to_Income, Assets_to_Loan, EMI_style (Section 3)
- Class weighting (class_weight='balanced') to address the 62/38 class imbalance
- Hyper-parameter tuning of Random Forest via GridSearchCV (5-fold CV, F1 scoring)
- Two independent feature-importance methods: built-in Gini importance and permutation importance on the held-out test set, to cross-check which features actually matter
- Ablation study: retrained the best model with the top feature (cibil_score) removed



Both importance methods agree: cibil_score dominates (Gini importance 0.7959, permutation importance 0.3438), with loan_term and the engineered Loan_to_Income ratio as distant secondary signals.

Ablation result: removing cibil_score and retraining dropped F1 from 0.9972 to 0.6616 -- a 0.3356 absolute drop (33.7% relative). This is direct evidence that the model's near-perfect accuracy rests almost entirely on one column, not a broad combination of signals.

8. Conclusion, Limits, and Future Work

Conclusion

Three classifiers were trained and compared using 5-fold cross-validated F1, test accuracy, precision, recall, F1, and ROC-AUC. The plain Decision Tree (max_depth=4) achieved the best raw test F1 (0.9991), slightly ahead of the tuned Random Forest (0.9972) -- tuning the ensemble added little because the problem is already close to linearly separable by cibil_score alone. Two independent importance methods and an ablation study confirm cibil_score is the dominant driver of loan approval in this dataset, with loan_term and Loan_to_Income as secondary factors.

Limits

- The dataset is very likely synthetic or heavily simplified -- a near-deterministic credit-score threshold is not how real-world bank underwriting works (real banks weigh many soft factors together).
- No time dimension in the data -- cannot tell if this reflects one bank's policy at one point in time, or changes over time.
- No formal fairness audit was performed. Approval rates were flat across education and self-employment groups in this dataset, which is reassuring, but a full audit (disparate impact testing, subgroup calibration) was out of scope.
- Possible proxy bias: income, assets, and employment status could correlate with protected characteristics not present in this dataset (e.g. gender, region, caste, religion) -- a real deployment would need that audit before use.

Future Work

- Try XGBoost / LightGBM for comparison against the scikit-learn ensemble baselines
- Apply SMOTE or other resampling techniques, even though imbalance here was mild
- Run a formal fairness/bias audit across any available demographic attributes
- Validate against real (non-synthetic) bank lending data if access becomes available

9. References

- Kaggle: Loan Approval Prediction Dataset (loan_approval_dataset.csv)
- scikit-learn documentation: https://scikit-learn.org/stable/documentation.html
- pandas documentation: https://pandas.pydata.org/docs/